AI.md

Artificial Intelligence Architecture

Project: Bloom

Document: AI System Design

Version: 1.0

Status: Core Intelligence Specification

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1. Purpose

Artificial Intelligence is the decision-making layer of Bloom.

Its purpose is **not** to replace teachers or generate educational content.

Its purpose is to understand learners, model their knowledge, identify skill gaps, and orchestrate personalized learning experiences using the best educational resources available across the internet.

AI is responsible for answering one fundamental question:

> **"Given who this learner wants to become and what they currently know, what is the best next learning experience?"**

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2. AI Philosophy

Bloom follows five AI principles.

Principle 1 — AI Understands Before It Recommends

Recommendations should never begin with content.

They begin with understanding the learner.

Principle 2 — AI Guides, It Does Not Replace Learning

Bloom does not solve problems for learners.

It guides them toward understanding.

Principle 3 — AI Optimizes Learning Outcomes

Traditional recommendation systems optimize:

- Watch time
- Clicks
- Engagement

Bloom optimizes:

- Skill acquisition
- Knowledge retention
- Career progression
- Concept mastery

Principle 4 — AI Must Be Explainable

Every recommendation should have a reason.

The learner should always understand:

- Why this resource?
- Why now?
- What comes next?

Principle 5 — AI Continuously Learns

Every interaction improves future recommendations.

Bloom's intelligence evolves alongside the learner.

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3. AI Architecture

```
                 User
                  │
                  ▼
         Learning Intent Engine
                  │
                  ▼
        Learner Knowledge Model
                  │
                  ▼
          Skill Gap Analyzer
                  │
                  ▼
       Learning Experience Planner
                  │
                  ▼
       Resource Retrieval Engine
                  │
                  ▼
     Recommendation Ranking Engine
                  │
                  ▼
          Personalized Learning Feed
                  │
                  ▼
          Learning Feedback Engine
                  │
                  └──────────────┐
                                 ▼
                 Updated Learner Model
```

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4. AI Components

4.1 Learning Intent Engine

Purpose

Transform human goals into structured learning objectives.

Example:

User Input

> "I want to become a Machine Learning Engineer."

The AI identifies:

- Career destination
- Existing knowledge
- Missing prerequisites
- Estimated timeline
- Learning preferences

Output:

A structured Learning Intent Profile.

4.2 Learner Knowledge Model

Bloom maintains a continuously evolving model of each learner.

Rather than storing completed videos, it models understanding.

The model estimates:

- Known concepts
- Weak concepts
- Unknown concepts
- Learning speed
- Preferred learning style
- Confidence level
- Career readiness

The model changes continuously as new signals arrive.

4.3 Skill Gap Analyzer

The analyzer compares:

Current Knowledge

↓

Target Career

↓

Required Skills

↓

Missing Skills

The output is an ordered list of learning priorities.

The learner never needs to manually determine what to learn next.

4.4 Learning Experience Planner

This component plans complete learning experiences.

Instead of recommending:

"Watch this video"

Bloom recommends:

```
Understand Concept

↓

Watch Explanation

↓

Read Documentation

↓

Practice

↓

Build Project

↓

Reflect
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↓

Review

↓

Advance
```

Learning becomes active rather than passive.

4.5 Resource Retrieval Engine

Responsible for discovering educational resources.

Searches across supported platforms.

Candidate resources may include:

- Videos
- Articles
- Documentation
- Projects
- Repositories
- Podcasts
- Interactive exercises

The retrieval engine returns candidates.

It does not rank them.

4.6 Recommendation Ranking Engine

Evaluates candidate resources.

Ranking considers:

- Relevance
- Difficulty
- Educational quality
- Creator suitability
- Learning style
- Skill dependency
- Freshness
- Previous learner interactions

Popularity alone is never sufficient.

4.7 Feedback Engine

Every learner interaction becomes feedback.

Examples:

Positive:

- Completed resource
- Saved resource
- Finished project
- Requested similar content

Negative:

- Abandoned resource
- Skipped recommendation
- Difficulty mismatch
- Repeated confusion

The learner model updates continuously.

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5. AI Decision Pipeline

Every recommendation follows the same reasoning process.

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Understand Learner

↓

Estimate Knowledge

↓

Identify Skill Gap

↓

Determine Next Skill

↓
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```
Retrieve Resources

↓

Evaluate Resources

↓

Build Learning Session

↓

Deliver Recommendation

↓

Observe Feedback

↓

Update Learner Model
```

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6. Learning Intent Profile

The Learning Intent Profile represents the learner.

Possible information includes:

Career Goal

Current Skills

Learning Preferences

Language

Daily Learning Time

Completed Skills

Weak Areas

Learning History

Career Timeline

Progress

The profile evolves continuously.

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7. Knowledge Representation

Bloom represents knowledge using interconnected graphs.

Different graph types include:

Skill Graph

Represents relationships between concepts.

Career Graph

Maps careers to required competencies.

Creator Graph

Models educators by:

- expertise
- teaching style
- explanation depth
- difficulty
- audience suitability

Content Graph

Represents educational resources and their relationships.

Learner Graph

Represents an individual's learning journey.

These graphs work together to generate recommendations.

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8. AI Memory

Bloom maintains long-term educational memory.

It remembers:

- Completed concepts
- Weak concepts
- Favorite creators
- Preferred explanation styles
- Previous mistakes
- Projects completed

This enables long-term personalization.

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9. Recommendation Strategy

Bloom recommends:

Skills

What should be learned.

Experiences

How it should be learned.

Resources

Where it should be learned.

Timing

When it should be learned.

Revision

When knowledge should be reinforced.

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10. Creator Intelligence

Bloom evaluates educators beyond popularity.

Possible dimensions include:

Teaching Style

Concept Depth

Practical Orientation

Visual Explanations

Project Focus

Interview Focus

Beginner Friendliness

Communication Clarity

Different learners benefit from different teaching styles.

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11. Content Intelligence

Bloom evaluates resources using multiple signals.

Examples:

Educational Quality

Accuracy

Freshness

Structure

Coverage

Difficulty

Prerequisite Alignment

Community Trust

Popularity is one signal among many.

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12. Adaptive Learning

The platform continuously adapts.

Example:

If the learner repeatedly abandons advanced resources:

AI may:

- reduce complexity
- recommend prerequisite concepts
- slow progression
- suggest alternative creators

The roadmap adjusts automatically.

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13. Explainable AI

Every recommendation should answer:

Why this?

Why now?

Which career objective does this support?

Which prerequisite does it satisfy?

What will I unlock afterward?

Explainability builds trust.

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14. Human Control

AI should assist, not control.

Learners may:

- ignore recommendations
- change career goals
- modify roadmaps
- disable Guard Mode
- choose alternative resources

AI remains advisory.

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15. Continuous Learning Loop

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Recommendation

↓

Learner Interaction

↓

Feedback Collection

↓

Learner Model Update

↓

Skill Graph Update

↓
```

```
Future Recommendation Improvement
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The system becomes increasingly personalized over time.

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16. Ethical Principles

Bloom's AI should:

- Respect learner autonomy.
- Minimize unnecessary data collection.
- Avoid manipulative engagement tactics.
- Be transparent about recommendations.
- Prioritize educational outcomes over platform usage.
- Never intentionally increase screen time for business purposes.

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17. Future AI Capabilities

Potential future capabilities include:

- AI career mentor
- Interview preparation assistant
- Personalized project generator
- Knowledge retention forecasting
- Multi-modal learning support
- Research paper simplification
- Team learning optimization
- Institutional curriculum intelligence

These features build upon the same learner model.

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18. Measuring AI Success

The AI system should be evaluated using learning-centered metrics rather than engagement metrics.

Examples include:

- Skill mastery progression
- Recommendation acceptance rate
- Learning path completion

- Knowledge retention
- Project completion rate
- Career milestone achievement
- User satisfaction with recommendations

The objective is not to maximize content consumption.

The objective is to maximize meaningful learning outcomes.

AI Summary

Bloom's AI is a **Learning Intelligence System**, not a chatbot or a traditional recommendation engine.

It combines learner understanding, knowledge modeling, skill graph reasoning, creator intelligence, content intelligence, and continuous feedback to orchestrate personalized learning experiences.

The long-term vision is an AI that understands **who a learner wants to become**, models **what they currently know**, predicts **what they should learn next**, and continuously guides them through the world's educational resources with clarity, transparency, and purpose.